How Australian Grocery Comparison Apps Get Their Prices, and What "Market List" Should Do

WiseList does not say publicly how it gets its prices, but everything it does say fits automated collection from the retailers' own websites. For Market List, my recommendation is to avoid scraping Coles and Woolworths from your own Vercel functions. Those requests come from data-centre addresses, which both chains block. Instead, pin the specific product IDs your household buys, refresh them once a week (Wednesday) through a paid scraping service that routes requests through Australian residential connections, and write the results into Firestore. For price history, use the free hotprices.org community dataset as a backup source.

TL;DR

- **WiseList and its peers:** WiseList's data source is **undocumented** (verified absence). What it does show is "live prices", postcode-based state pricing, barcode scan, and cart transfer to Coles and Woolworths. Its founder has also admitted the "engine" once misread a Woolworths member-price tag as a shelf price. Together these strongly suggest it reads retailer web data (inferred, not confirmed). Frugl is the one peer with some documentation: it compiled "publicly available sales info" and sold shopper-behaviour data to Coles. Open-source trackers such as hotprices-au openly scrape Coles and Woolworths daily.
- **What is technically available in October 2026:** There is still no official retailer API.
  - **Woolworths:** undocumented JSON endpoints (for example `POST /apis/ui/Search/products`) behind Akamai.
  - **Coles:** Next.js `/_next/data/<buildId>/…json` routes behind Imperva/Incapsula.
  - **Aldi:** publishes a national catalogue of roughly 3,200–3,300 products with prices but no online ordering.
  - **Blocking and policy:** Both majors block data-centre IP addresses. Government policy is moving towards *guaranteeing* that third parties can scrape, not towards an API, but nothing has been legislated yet.
- **What to build:** a pinned-item weekly job, run through a paid scraping service, feeding Firestore. Apify's rel8ble Coles actor charges "$1.50 per 1,000 results", and clearpath's Coles actor charges "$1.49 per 1,000 rows (Free plan) down to $0.99 (Gold and above)". Add hotprices.org history as a free supplement.
  - This reverses if any of these happen: the government mandates scraping access or an API (then fetch directly), the app goes commercial (licensing and terms risk rises), or

the paid actors break for an extended period (fall back to manual or receipt entry).

Key Findings

1. WiseList: what is known vs inferred vs unknown

Verified from WiseList's own materials:

- **Coverage:** Coles, Woolworths and ALDI. The App Store listing says WiseList "compares Coles, Woolworths and ALDI as you add each item, then shows you the cheapest way to shop this week." apple apple The website claims "Live prices from Coles, Woolworths & ALDI" and "400,000+ Australians signed up." wiselist
- **Regional pricing:** "Prices and specials differ from state to state. WiseList uses your postcode to show the ones for your area." wiselist This implies state-level (not individual-store) price sets.
- **ALDI matching:** "ALDI sells many products under its own brands, so WiseList compares an ALDI product only where it can match an equivalent." wiselist wiselist An older help page says only "perishable items like milk, fruit & vegetables" were comparable across all three. wiselist
- **Basket totals:** "the total price is the sum of comparable items—not the total price of your whole grocery list." wiselist An older help article says the total shows $0 when nothing on the list has comparable prices at both Coles and Woolworths, "to avoid misleading comparisons." wiselist wiselist
- **Split by Store:** the paid tier ($7.99/mo or $49.99/yr) adds "Split by Store", Scan & Compare by barcode, and cart transfer "to Coles or Woolworths and arrange delivery or click and collect with them." apple +3
- **Basket Test:** WiseList runs a monthly "WiseList Basket Test", a fixed 34-item basket priced at all three majors across seven capital cities. Founder Adrian Liu posted on LinkedIn that Coles and Woolworths matched "to the cent" on 55% of comparisons in August, falling to 47% in September. linkedin
- **The member-price error:** In the same LinkedIn thread Liu disclosed that "the two item prices I named carried a Woolworths member-price tag our engine read as a shelf price." linkedin

Inferred (not confirmed):

- An "engine" that reads retailer-applied price tags (member price vs shelf price), plus "live" prices by state, is most consistent with automated collection of the retailers' own online product data. It is not consistent with catalogue parsing or crowdsourcing.
- Barcode lookup implies a product database keyed by GTIN/EAN. Woolworths exposes GTINs publicly and Coles exposes them on product pages. apify dev

Unknown:

- No founder interview, privacy policy excerpt, job ad or forum thread I found states the source: scraping, licensed feed, partnership or anything else.
- Its investor, Galileo Ventures, describes it only as a comparison app founded by software engineer Adrian Liu, with seed investment in 2021. galileo galileo
- WiseList's privacy policy and terms pages exist, but I did not verify them. Do not assume a data partnership with Coles or Woolworths; I found no evidence of one.

2. Other Australian apps and open-source trackers

Project	How it gets data	Evidence level
Frugl (Family Insights Group, ASX: FAM)	Compiles "publicly available sales info"; claims 58,280 products across 5,266 stores, with Coles/Woolworths/ALDI/IGA/Drakes updated "every week". Revenue came from selling behavioural data: founder Ramzi Abboud said "Selling data to Coles is a substantiation of our business model." stockhead +2	Verified (press + website); collection mechanism not specified
Grocerize	Compares Coles vs Woolworths, with price history "growing each week" and "New data is added every time prices change." grocerize	Collection method undocumented; inferred web collection
hotprices.org / Javex/hotprices-au (MIT)	Fork of Austria's heissepreise; "fetches data from the duopoly of Australian grocery chains on a daily basis." Python scraper plus static frontend. github	Verified (README)
Price Hipster	Crawled retailer sites; Woolworths data disappeared in 2021–2022. ozbargain Its operator said "Woolworths' servers are being overly defensive" and that no one at Woolworths would engage.	Verified (OzBargain store-rep posts)
	23 pinned products, fetched weekly from each chain's	Verified

BRACE777/grocery-prices	undocumented endpoints via curl on a *home PC*; history from hotprices.org. github	(README)
tjhowse/aus_grocery_price_database (auscost)	Go service that "reads from grocery store web APIs" (Woolworths and Coles) into InfluxDB/Grafana. github	Verified (README)
Grocermatic	Open-source scraper and web app tracking Coles and Woolworths. github	Verified (README)
arkind13/grocery-price-tracker	Says its live web source was "RETIRED 2026-09-02 … after the lost store-bot war"; now uses documents as input. github	Snippet only; treat as indicative
Half Price, Salefinder/Lasoo, OzBargain trackers	Not investigated in depth within this research budget.	Unknown

What the evidence shows: Every open-source tracker scrapes. The one commercial app with a documented business model (Frugl) monetised data *outward* to retailers rather than licensing it *inward*. No Australian comparison app I found claims an official retailer data feed.

3. Technical data sources as of late 2026

Woolworths

- **Endpoints:** Undocumented JSON endpoints that the site itself uses. An October 2026 GitHub issue confirms `POST /apis/ui/Search/products` works on the live site. `GET /apis/ui/browse/category?…` with query-string filters is "likely broken" because category browse is now a JSON POST, and `GET /apis/ui/product/detail/{stockcode}` is unverified. github
- **Bot protection:** Akamai Bot Manager, which "fingerprints the TLS handshake and rejects generic scrapers with Access Denied." apify
- **Data-centre blocking:** One hobbyist found "Woolworths returns 403 to data centre addresses. The same request from a home connection returns 200." github
- **History of blocking:** Woolworths has blocked trackers before: Price Hipster (2021–2022), ozbargain and an OzBargain user's script calling `apis/ui/product/detail` hit "security checks" in 2021. ozbargain

Coles

- **Endpoints:** A Next.js site. Every search, category and product page has a matching

`/_next/data/<buildId>/…json` route, and the buildId changes with every deploy. apify +2

- **Bot protection:** Imperva/Incapsula. Coles "blocks datacenter IP addresses", and blocked pages can come back as HTTP 200 with a tiny challenge body, so failures can be silent. apify scrapfly
- **Header quirk:** BRACE777 found Coles served a bot challenge to requests with `Accept-Language: en-AU,en;q=0.9` but the real site with `en-US`. It also found Python's HTTP client was challenged where curl wasn't. github
- **Durability:** These details are fragile and should be treated as perishable. github

ALDI Australia

- **Catalogue:** ALDI states it "does not currently offer online ordering of groceries and the ALDI website serves as an online grocery catalogue of all everyday products and seasonal Limited Time Only and Special Buys products." Product pages show shelf price and unit price; one scraper reports about 3,300 products in the full range. aldi apify
- **No store-level prices:** Per-store catalogues are partial; one store returned 713 of about 3,200 products. So there is effectively a *national* price only. apify
- **Conflicting statement:** The ALDI homepage says "ALDI's website only displays current prices for reduced items." aldi This conflicts with product pages that do list everyday prices. Verify on the live site before relying on it.

Catalogues (Salefinder, Lasoo, retailer PDFs): Not researched in depth. They cover only advertised specials, not the full range. Frugl's CEO noted each major runs "roughly 3000-3500 items on promotion each week", far more than any printed catalogue shows. yahoo

Store and region pricing:

- WiseList prices by state (postcode). wiselist
- BRACE777 found Woolworths offers no store selection without signing in, and two Victorian Coles stores returned identical prices. Its conclusion: "Prices are national online prices and in-store may differ." github
- Some Coles scraper services accept a `storeId`. apify
- Member prices (Everyday Rewards, Flybuys) "are [not] published without signing in", which BRACE777 confirmed across 250 Woolworths and 53 Coles products. github

4. Official and commercial data options

- **No retailer API:** There is no public Coles or Woolworths API, and no affiliate data feed was found. scrapingbee +2
- **ACCC Supermarkets Inquiry, final report (February 2025), Recommendation 2 (p. 114):** large chains should publish prices online. It also calls for "very large supermarket

chains (by turnover) to make application programming interfaces available which provide dynamic price information for third parties."

- **Treasury consultation, "Improving price transparency in the supermarkets industry":** Treasury's Consult hub (c2026-732517) says it opened 19 January 2026 and closed 17 February 2026; Treasury's main consultations list gives the dates as 20 January – 17 February 2026. It sought views on requiring supermarkets to "publish prices in-store, publish prices online, enable web-scraping technologies, display minimum information about promotions, provide loyalty program information every 6 months." Andrew Leigh's media release of 20 January 2026 described the proposal as requiring large supermarkets to "publish prices online, ensure web-scraping technologies can be used by third parties like online comparison websites and apps".
  - The government material reviewed does not mention the ACCC's API proposal.
  - CHOICE's February 2026 submission said the government "should give further consideration to… developing a supermarket pricing API".
  - The Competition and Consumer Law Committee of the Law Council's Business Law Section, in a submission dated 17 February 2026, urged Treasury to "adopt a web-scraping-based model for price comparison tools and avoid mandating APIs, which would impose a significant compliance burden on supermarkets."
- **Status as of October 2026:** No exposure draft or start date was found. Leigh's 27 June 2026 release still lists price transparency among reforms being implemented. treasury Treat it as **pending**.
- **Unit Pricing Code:** consulted on 1–19 September 2025 (scope, display, cross-retailer consistency, shrinkflation, civil penalties). treasury No finalised amendment was found.
- **Excessive pricing ban:** in force from 1 July 2026 under the Food and Grocery Code. Very large retailers must keep pricing information for three years, addisons claytonutz but this is a record-keeping duty to the regulator, not a public feed.
- **Commercial options:**
  - **Scraping-as-a-service:** Apify actors such as rel8ble's Coles scraper ("$1.50 per 1,000 results"), clearpath's Coles scraper ("$1.49 per 1,000 rows (Free plan) down to $0.99 (Gold and above)" plus "$0.005 per run start") and parsebird's ALDI scraper ("Free $1.99, Bronze $1.79, Silver $1.59, Gold $1.29" per 1,000), as well as ScrapingBee and Parse.bot's "managed, monitored" Woolworths API.
  - **Bespoke data scrapers:** Actowiz, FoodDataScrape. actowizsolutions fooddatascrape
  - **Who actually holds the data:** These are resellers of scraped data, not licensed data. NielsenIQ/Circana-type retail panels exist, but I found no evidence they offer affordable shelf-price feeds for a hobby app (unknown).

5. Legal and terms of service (not legal advice)

- **Fact:** Woolworths' online terms say users must not "use any robot, spider, site search and retrieval application or other mechanism to retrieve or index any portion of the Site." woolworths I did not retrieve Coles' terms (unknown), but assume similar wording.
- **Fact:** Australian copyright does not protect facts as such. *IceTV v Nine Network* (High Court, 2009) and *Telstra v Phone Directories* (Full Federal Court, 2010) set the rule, and Australia has no EU-style database right. lexology lexology Prices are facts. Product images and marketing descriptions are likely copyright works, so don't copy them into your app.
- **Fact:** I found no lawsuits or published cease-and-desist letters against Australian grocery comparison apps. Enforcement so far has been *technical* (IP blocking, as with Price Hipster). ozbargain +2
- **Opinion/inference:**
  - The main exposure is breach of contract under the terms of service. Enforceability is uncertain where terms are browse-wrap. eversheds-sutherland
  - Getting around bot protection with residential proxies increases the risk, ethically and possibly legally. aaronhall
  - For a private household tool fetching about 50 items weekly, practical risk is very low. It rises sharply if you publish data, go commercial, or hammer endpoints.
  - The pending Treasury proposal would, if enacted, *require* the majors to tolerate third-party scraping. andrewleigh
- **Privacy/Spam:** Scraping product prices involves no personal information. Your Firestore household data is what falls under privacy obligations.

6. Product matching and basket logic

- **Identifiers:** Pin each list item to retailer-specific product IDs (Woolworths stockcode, Coles product ID, ALDI product ID), which is the approach BRACE777 takes ("No guessed products"). github Use GTIN/EAN where exposed to auto-match branded goods across chains.
- **House brands (Coles, Woolworths, ALDI exclusives):** These have no shared barcode. Equivalence has to be judged by a human or a rule (same category, comparable size and grade). WiseList itself limits ALDI comparisons to items with genuine equivalents. wiselist wiselist
- **Unit price:** Normalise to $/kg, $/L, $/100g or $/each, and compare on unit price. "A larger pack can win while costing more." github Handle loose produce carefully: ALDI shows both a per-kg price and an estimated per-piece price ("Garlic is $29.99/kg but $1.80 a bulb"). apify
- **Basket rules:**
  - Compute (a) a single-store total over *comparable items only* (WiseList's rule) and (b) a

cheapest-per-item split. wiselist wiselist
    - Show the split's savings net of the hassle of a second trip.
    - Flag items missing at a store instead of silently excluding them.
- **Specials:**
    - Store the current price, the was-price, the special flag and the multibuy text ("Pick any 2 for $44"). dev
    - An ABC News analysis (26 October 2024) of online prices for "nearly 44,000 items" between late May and mid-August 2024 found "roughly 2,500 products moved in weekly cycles, mostly alternating between two or three price points" (48% Woolworths, 52% Coles), so a "special" may just be the low half of a cycle.
    - Rank deals by how unusual the price is against its own 90-day history, as BRACE777 does, rather than by the advertised discount. github
- **Member prices:** Exclude them, or label them separately. They aren't visible without login, and WiseList's misreading shows how easily they contaminate shelf-price comparisons. github linkedin

7. Update frequency and infrastructure

- **Weekly cycle:** Specials change on Wednesdays, so a weekly run shortly after Wednesday morning captures most changes. BRACE777 fetches only when its data predates "the most recent Wednesday 7am." github dev
- **Catalogue size:** About 44,600 Coles and Woolworths products in hotprices data since September 2023, github and about 3,300 ALDI products. A full catalogue scrape means tens of thousands of requests per run against hostile bot protection. That makes no sense for a household app.
- **Pinned-item cost:** 60 items × 3 retailers × about 4.3 weeks ≈ 800 results a month, which comes to about $1 a month at published Apify per-result rates (an estimate; platform minimums or subscription tiers may apply).
- **hotprices.org data:** Independent projects report hotprices.org publishes daily gzipped JSON per store. The files hold price-change history only (no was-price or special flag), and the licence is unclear. One developer obtained the maintainer's permission by email. github github The exact URLs were not verified.

Recommendations: options compared

Option	Effort	Fragility	Cost	Legal/ToS risk	Verdict
A. Vercel cron		**Very high**: data-centre			

fetching Coles/Woolworths directly	Low–medium	IPs blocked (Woolworths 403; Coles Imperva) github	Free	Medium (ToS)	Won't work reliably
B. Pinned items via paid scraping service (Apify) → Vercel webhook → Firestore	Medium (1–2 weekends from a phone)	Medium: vendor maintains anti-bot workarounds; actors can break	About $1–5/mo (estimate)	Medium (ToS; proxy circumvention)	**Recommended**
C. Reuse hotprices.org daily dump (filtered by GitHub Action/Vercel cron)	Low	Medium: depends on a volunteer; national prices; no special flags github	Free	Low–medium; ask maintainer permission	**Best supplement** (history and fallback)
D. Full-catalogue scraper of your own	High	High	Proxy costs grow with volume	Higher	Not justified
E. Manual or receipt entry (WiseList offers receipt scan) wiselist	Low	None	Free	None	Fallback
F. Licensed commercial data	Unknown	Low	Likely enterprise pricing	Low	Only if commercial

Suggested build for Market List:

1. Add a `products` collection in Firestore with `{woolworthsId, colesId, aldiId, gtin, unitBase, sizeQty}` per list item. Add pins by pasting retailer product URLs, as BRACE777's `add_item` does. github
2. Schedule a Vercel cron (or the paid service's own scheduler) for Wednesday mid-morning AEST. Have it call the paid actors with only your pinned IDs and write `{price, wasPrice, unitPrice, onSpecial, offerText, fetchedAt}` to `prices/{productId}/{date}`.

3. For ALDI, try a direct fetch of ALDI product pages first (there is no evidence of ALDI data-centre blocking, but this is unverified). Fall back to a paid ALDI actor.
4. Optionally run a daily GitHub Action that pulls hotprices.org's dump, filters to your pinned IDs, and stores history. Ask the maintainer first.
5. In the PWA, compute single-store totals over comparable items, the best split, and a "genuinely low" flag against 90-day history. Show member prices separately or not at all.

When this recommendation reverses:

- If Treasury's scraping-access rule (or an API) is enacted, switch B to direct fetching.
- If Market List becomes public or commercial, get legal advice and consider licensed data.
- If paid actors fail for weeks, lean on C plus manual entry.

Caveats

- WiseList's actual pipeline is unknown; the inference rests on indirect clues.
- Endpoint and anti-bot details come mostly from scraper vendors and hobbyist READMEs (self-interested or anecdotal) and change frequently.
- Coles' terms, Half Price, Salefinder/Lasoo and Costco/IGA/Harris Farm sources were not verified.
- Cost figures are estimates from published per-result rates.